

ARMAN HEIDARI

Tehran, Iran • armanheidari192@gmail.com • +98 930 903 9003

[LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [ORCID](#) | [Personal Portfolio](#)

RESEARCH INTERESTS

- Deep Reinforcement Learning
- Representation Learning & Deep Generative Models
- Out-of-Distribution & Anomaly Detection
- Algorithmic Foundations & Optimization in DL

EDUCATION

Sharif University of Technology

Master of Science in Artificial Intelligence

Tehran, Iran

Expected: 2028

Allameh Tabataba'i University

Bachelor of Science in Computer Engineering

Tehran, Iran

Sep 2021 – Jul 2025

- **Cumulative GPA:** 19.29 / 20.00 (Ranked 1st among Class of 2025 graduates)
- **Selected Coursework:** Linear Algebra (20/20), Probability & Statistics (19/20), Artificial Neural Networks (20/20), Natural Language Processing (20/20), Artificial Intelligence (20/20), Data Structures & Algorithms (20/20), Design & Analysis of Algorithms (17.5/20).

HONORS & AWARDS

- **Ranked 1st in the Nationwide M.Sc. University Entrance Exam (Konkur)** among 20,000+ participants in the Computer Engineering group, 2026.
- **Ranked 1st Academic Standing** among all undergraduate Computer Engineering students, Allameh Tabataba'i University, Class of 2025.
- **7th Place, ACM-ICPC Regional Collegiate Programming Contest**, demonstrating advanced algorithmic design, dynamic programming, and rapid problem-solving, 2022.

TEACHING EXPERIENCE

Department of Computer Engineering, Allameh Tabataba'i University — Teaching Assistant

- **Fall 2024:** Dr. Zahra Rashidi (Fundamentals of Programming – C/C++), Dr. Hassan Rashidi (Programming Languages), Dr. Karimi (Introduction to Databases)
- **Fall 2023:** Dr. Zahra Rahimi (Data Structures and Algorithms), Dr. Mobasheri (Fundamentals of Programming – Python), Dr. Ghaemi (Signals and Systems)
- **Fall 2022:** Dr. Shokouh Shahbeyk (Engineering Mathematics)
- **Responsibilities:** Conducted weekly discussion sections, designed algorithmic and coding assignments, held office hours, and graded assignments for classes of 30–40 undergraduate students.

PUBLICATIONS & PREPRINTS

S. Jani, **A. Heidari**, A. Anvari, and Z. Rahimi, "Intelligent Scientific Literature Explorer using Machine Learning (ISLE)," *arXiv preprint arXiv:2512.12760 [cs.LG, cs.AI, cs.CL]*, Dec 2025. Supervised by Dr. Zahra Rahimi.

- Architected a scientific discovery engine integrating hybrid retrieval (BM25 + dense sentence embeddings via Reciprocal Rank Fusion) and automated semantic clustering (BERTopic/NMF).
- Engineered a heterogeneous knowledge graph mapping multi-hop entity relationships and thematic evolution across arXiv and OpenAlex datasets.

WORK EXPERIENCES

Machine Learning Engineer | Jul 2024 – Aug 2025

Pars System Energy

➤ Industrial Predictive Maintenance & Anomaly Detection System

- **System Architecture:** Developed a real-time anomaly detection pipeline for multi-channel industrial IoT telemetry (e.g., chillers), combining deep generative modeling and sequential state forecasting.
- **Machine Learning Methodology:** Deployed an Adversarially Learned Anomaly Detection (ALGAN: Adjusted-LSTM GAN) framework alongside deep neural predictors to capture nominal dynamics and isolate operational anomalies.
- **Automated Decision Engine:** Integrated an automated Genetic Algorithm (GA) to synthesize and optimize fuzzy inference rules, dynamically assigning anomaly severity scores without manual heuristic tuning.
- **Operational Impact:** Substantially mitigated false alarm rates, lowered diagnostic latency, and prevented abnormal energy spikes caused by suboptimal operational degradation.

➤ Autonomous Multi-Agent Customer Support & Monitoring Ecosystem

- **Multi-Agent Orchestration:** Architected a 9-agent collaborative workflow using LangChain and Llama-3.1 to manage multi-turn dialogues, real-time intent dispatching, and automated issue resolution.
- **Context Retrieval & Reflection:** Implemented dense Retrieval-Augmented Generation (RAG) using FAISS vector indexing; incorporated a reflective agent to inspect, verify, and polish responses prior to transmission.
- **Safety & Telemetry Guardrails:** Constructed topic boundary enforcement and behavioral moderation guardrails alongside an extractor agent streaming categorized customer issue telemetry to central monitoring.

TECHNICAL SKILLS

Programming Languages:	Python, C++, C, SQL
Machine Learning & Deep Learning:	PyTorch, TensorFlow, Scikit-learn, Hugging Face
Agentic AI & Retrieval Systems:	LangChain, FAISS, Sentence-Transformers, RAG pipelines
Data Analysis & Scientific Computing:	NumPy, Pandas, Matplotlib, Seaborn
Developer Tools & Environments:	Git, Docker

ACADEMIC REFERENCES

Available upon request.